TikZ AI Plans Library — User Guide

A TikZ library for drawing AI partial-order planning diagrams: action nodes with precondition/effect label lines, init/goal bars, and causal links.

You will need the following files:

- `tikzlibraryaiplans.code.tex` (library code)
- `main.tex` (the main document)
- `domain.tex` (for defining the action schemas)
- `plan.tex` (the plan that uses the defined actions)

Loading the library

When the library file is in your project folder:

```
\usetikzlibrary{aiplans}
```

When the library file is in a different folder (e.g. two levels up):

```
\input{../../LibraryCode/tikzlibraryaiplans.code.tex}
```

Also add the required TikZ shape libraries for your document if needed:

```
\usepackage{tikz}
\usetikzlibrary{positioning, shapes.symbols, arrows.meta}
```

Overview of node types

Style	Description
`action={precs}{effs}{name}`	Rounded rectangle action node
`init={labels}`	Vertical black bar — horizontal layout init
`goal={labels}`	Vertical black bar — horizontal layout goal
`init vertical={labels}`	Horizontal black bar — vertical layout init
`goal vertical={labels}`	Horizontal black bar — vertical layout goal

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1. Action nodes

Basic syntax

```
\node[action={prec1,prec2}{eff1,eff2}{Action Name},minimum height=1cm] (a) {};
```

- **Argument 1** — comma-separated precondition labels (empty space for none)
- **Argument 2** — comma-separated effect labels (empty space for none)
- **Argument 3** — the action name shown inside the node



Named anchors

After a node is placed, the library creates named coordinates for every label line tip:

- `(nodename-pre-1)`, `(nodename-pre-2)`, … — precondition line tips
- `(nodename-eff-1)`, `(nodename-eff-2)`, … — effect line tips

These are the attachment points for causal links.

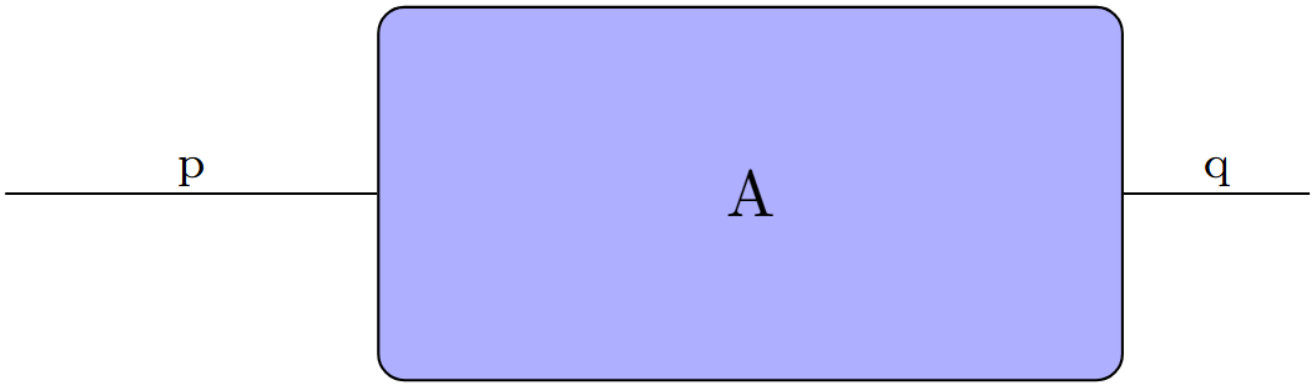
Default appearance

- Rounded corners, cyan fill, `minimum width=3cm`
- Label lines auto-sized to fit their text (widest label + 0.3 cm padding)
- Labels sit at the **midpoint** of each line, above it (horizontal) or to the right (vertical)

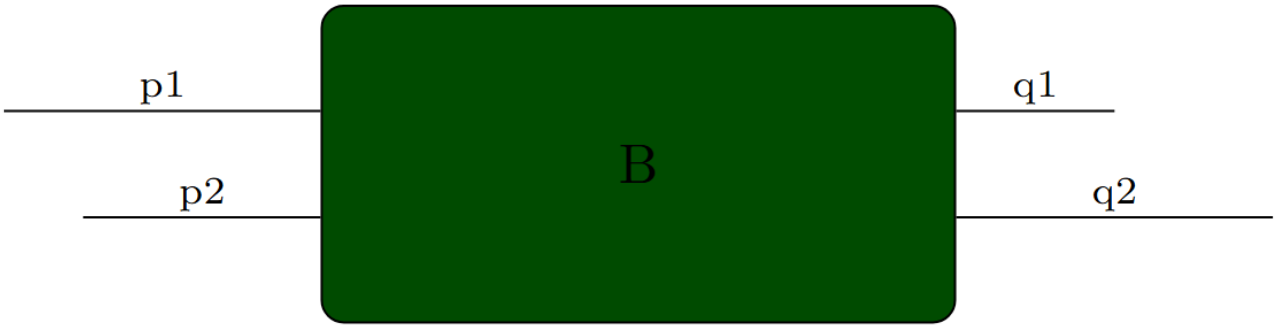
Overriding node size

Pass standard TikZ keys alongside the style:

```
\node[action={p}{q}{A}, minimum width=4cm, minimum height=2cm, fill=blue!30] (a)
{};
```




```
 \node[action={p1,p2}{q1,q2}{B}, minimum width=4cm, minimum height=2cm, fill=
{rgb,255:red,0; green,76; blue,0}, prec label lengths={2cm,1.5cm}, eff label
lengths={1cm,2cm}] (a) {};
```



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2. Layout: horizontal vs vertical

Horizontal (default)

Precondition lines extend **left** from the west side; effect lines extend **right** from the east side.

```
\node[action={p}{r}{MyAction}, minimum height=0.5cm] (a) {};
```

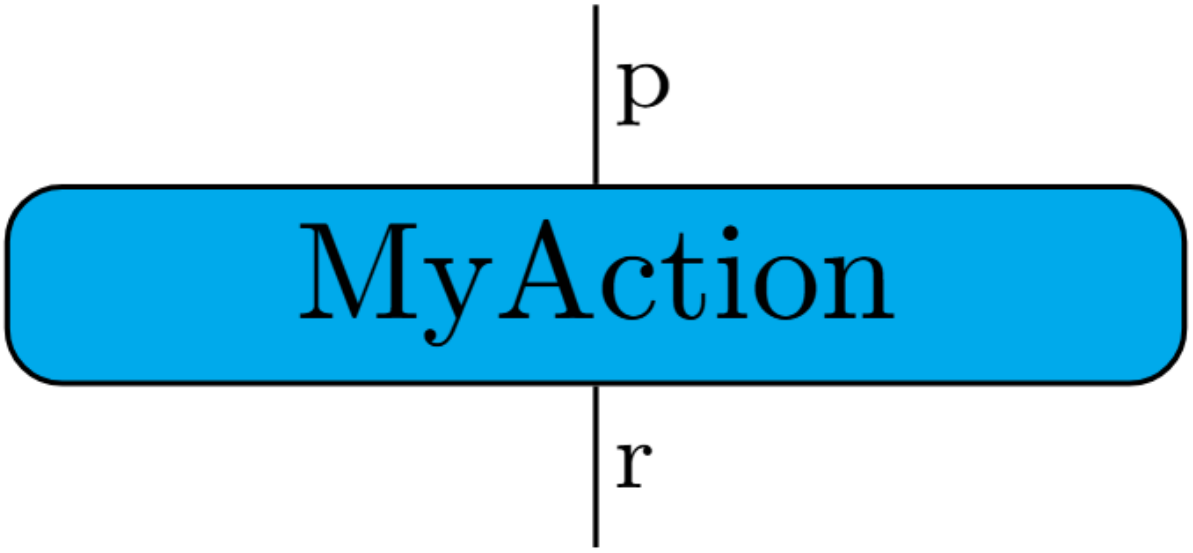


Vertical

Precondition lines extend **upward** from the top; effect lines extend **downward** from the bottom.

Add the `vertical` key anywhere in the node options:

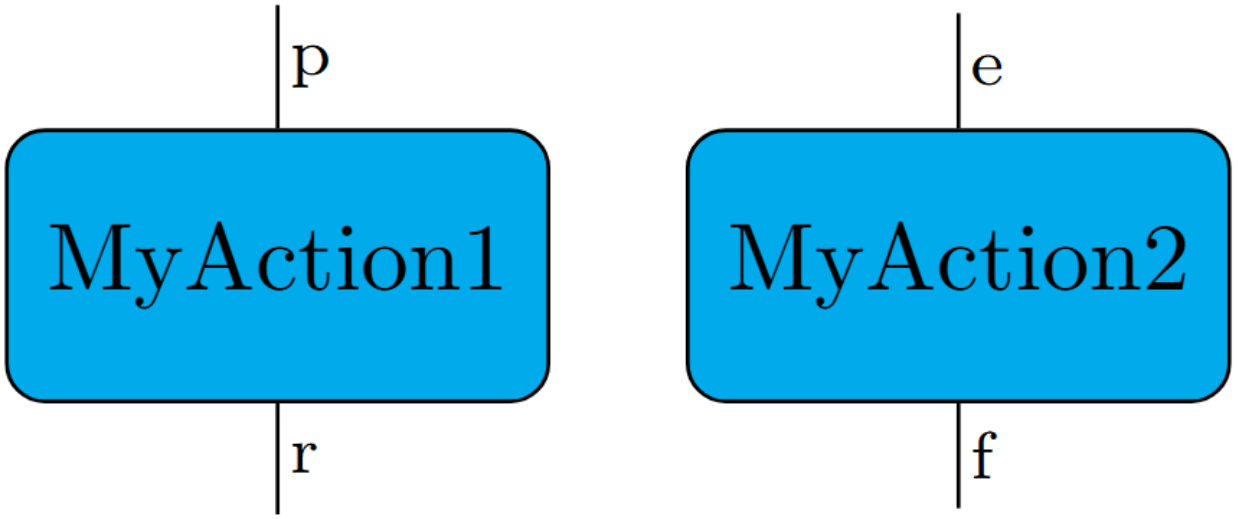
```
\node[action={p}{r}{MyAction}, vertical, minimum height=0.5cm] (a) {};
```



You can bake it into a shared style (also you could baked into shared style for horizontal style see Section 9 example):

```
\tikzset{
    VSTYLE/.style n args={3}{
        action={#1}{#2}{#3},
        vertical, minimum width=2cm, minimum height=1cm
    }
}
```

```
\node[VSTYLE={p}{r}{MyAction1}] (a) {};
\node[VSTYLE={e}{f}{MyAction2}] (b) [right=0.5cm of a] {};
```



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3. Init and goal nodes

Horizontal layout — `init` and `goal`

Tall thin vertical bars. Effect lines go **right** from init; precondition lines go **left** from goal.

```
\node[init={p,q,r}] (init) {};
\node[goal={$\neg$ p,$\neg$ q,$\neg$ r}] (goal) [right=10cm of init] {};
```



Default size: `minimum height=8cm`. Override freely:

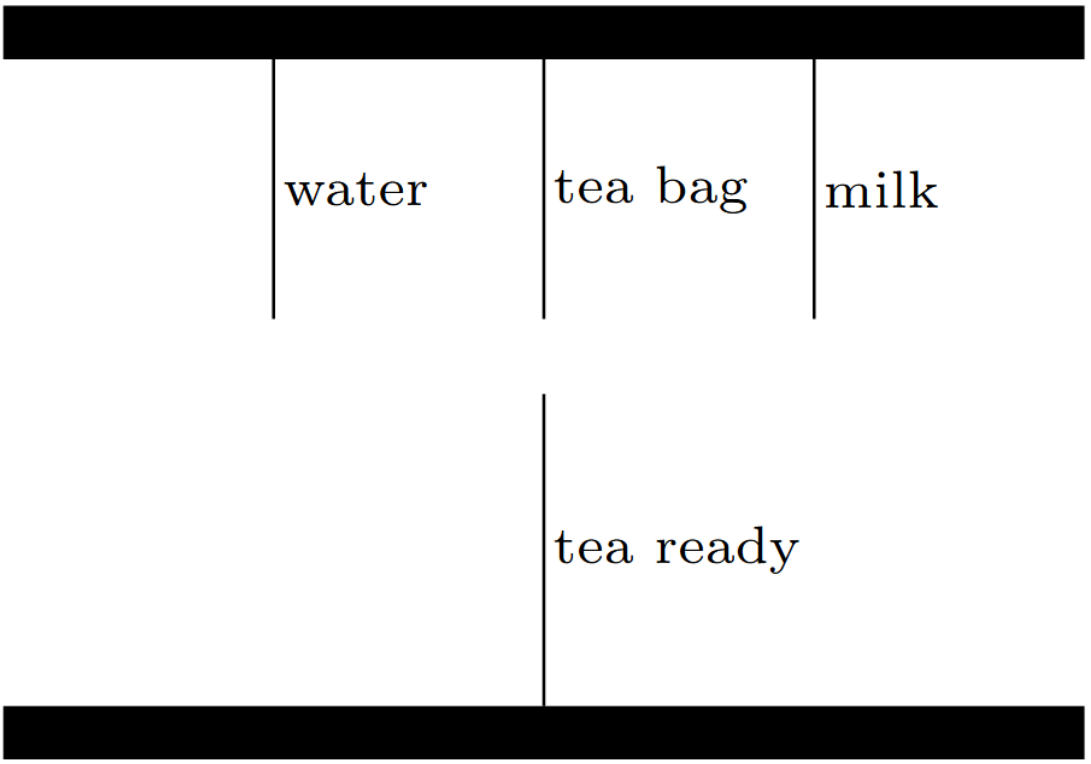
```
\node[init={p,q,r}, minimum height=4cm] (init) {};
\node[goal={$\neg$ p,$\neg$ q,$\neg$ r},minimum height=4cm] (goal) [right=10cm of
init] {};
```



Vertical layout — `init vertical` and `goal vertical`

Wide thin horizontal bars. Effect lines go **downward** from init; precondition lines go **upward** from goal. Default size: `minimum width=3cm`.

```
\node[init vertical={water,tea bag,milk}, minimum width=5cm] (init) {};
\node[goal vertical={tea ready}, minimum width=5cm] (goal) [below=8cm of init] {};
```



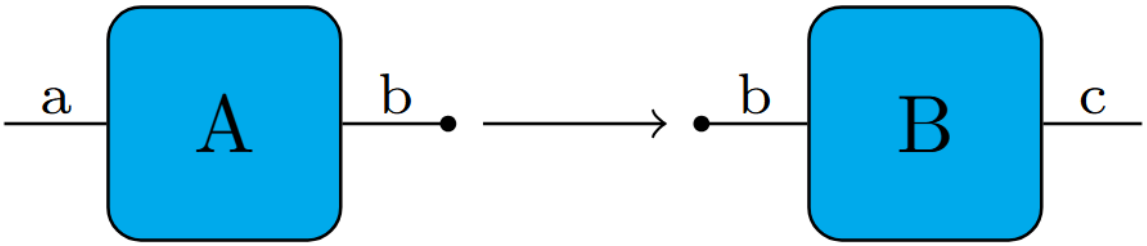
---

4. Causal links

Basic causal link

```
\node[action={a}{b}{A}, minimum width=1cm, minimum height=1cm] (a) {};
\node[action={b}{c}{B}, minimum width=1cm, minimum height=1cm] (b) [right=2 cm of
a]{};
\draw[causalLink] (a-eff-1) to (b-pre-1);
```

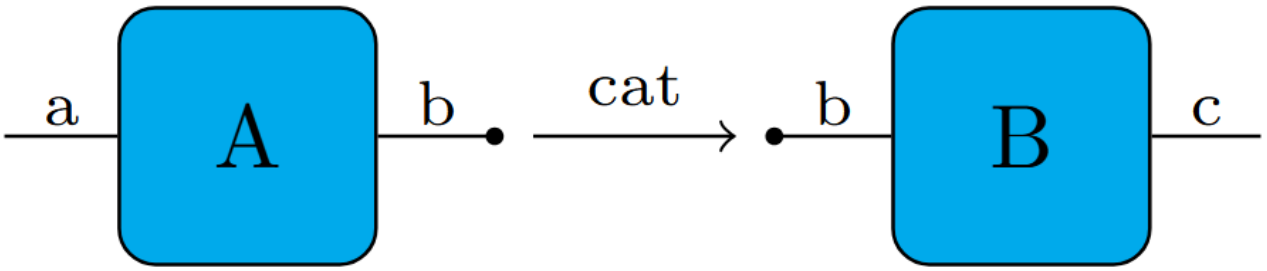
Draws an arrow with filled-circle endpoint marks at both ends, shortened slightly from each anchor.



Causal link with a mid-path label

```
\draw[causalLink with label={cat}] (a-eff-1) to (b-pre-1);
```

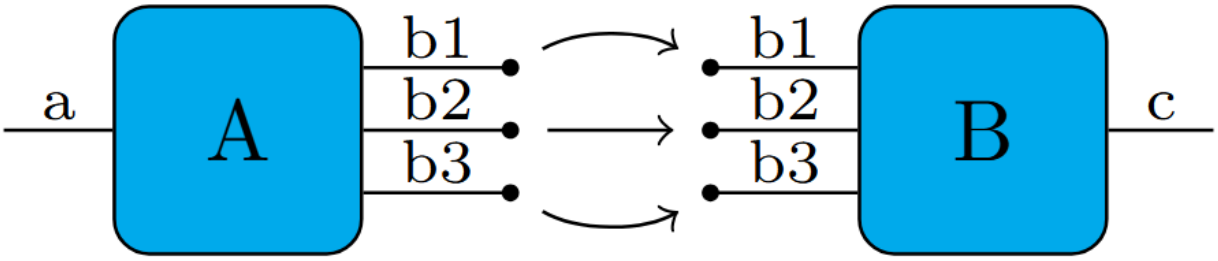
Places a small label above the midpoint of the link.



Bending and routing

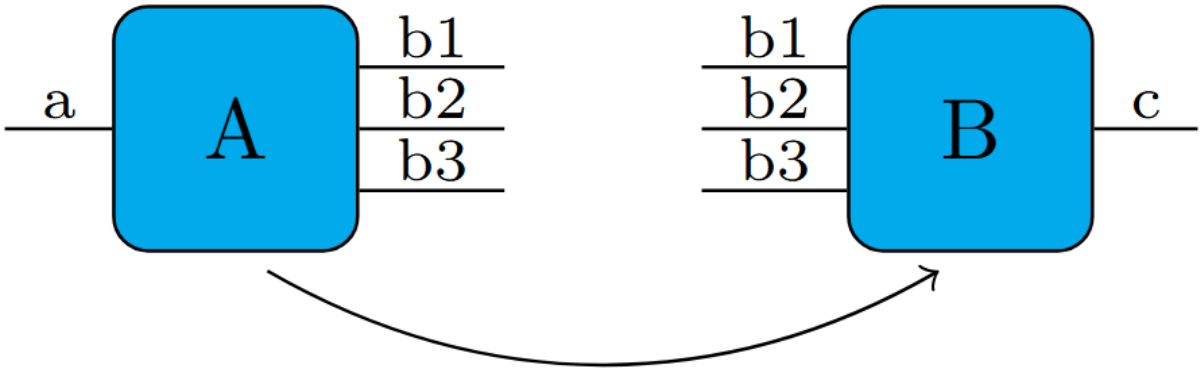
Standard TikZ path options work normally:

```
\node[action={a}{b1,b2,b3}{A}, minimum width=1cm, minimum height=1cm] (a) {};
\node[action={b1,b2,b3}{c}{B}, minimum width=1cm, minimum height=1cm] (b) [right=2
cm of a]{};
\draw[causalLink, bend left] (a-eff-1) to (b-pre-1);
\draw[causalLink] (a-eff-2) to (b-pre-2);
\draw[causalLink, bend right] (a-eff-3) to (b-pre-3);
```



You can also connect directly to a whole node rather than a specific anchor:

```
\draw[causalLink, bend right, causal mark=-] (a.south) to (b.south);
```



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5. Causal link endpoint marks

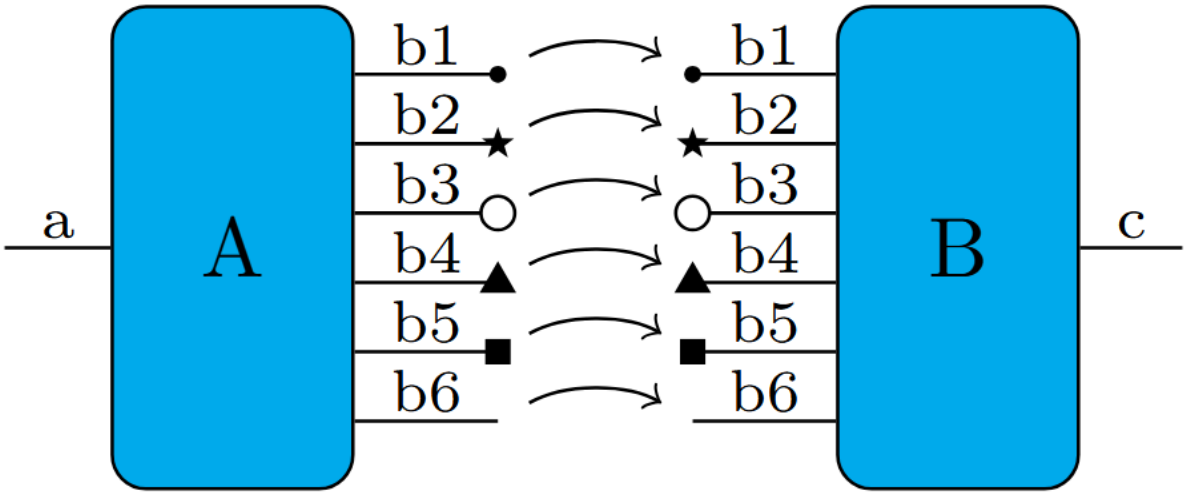
By default every causal link draws a **filled circle** at both ends. Override with `causal mark=X` placed **after** `causalLink` or `causalLink with label`:

Key	Mark
(default / omitted)	Filled circle
`causal mark=*`	Filled 5-pointed star
`causal mark=@`	Hollow circle

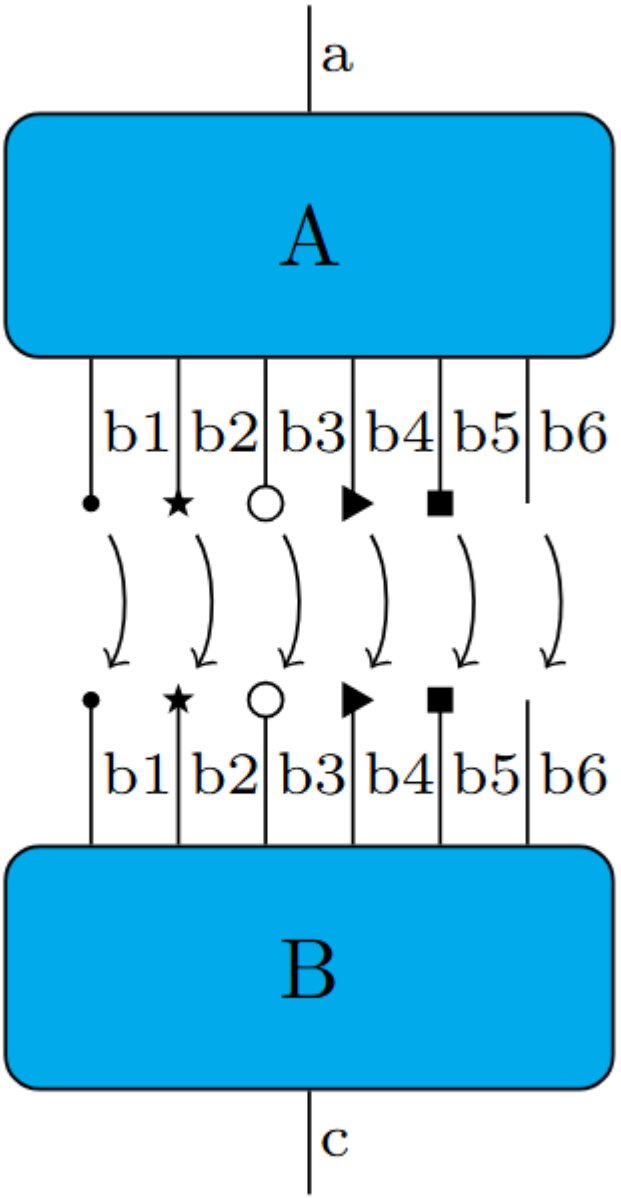
Key	Mark
`causal mark=^`	Filled triangle (direction-aware)
`causal mark={&}`	Filled square (braces required around `&`)
`causal mark=-`	No mark (plain arrow only)

```
\node[action={a}{b1,b2,b3,b4,b5,b6}{A}, minimum width=1cm, minimum height=2cm] (a)
{};
\node[action={b1,b2,b3,b4,b5,b6}{c}{B}, minimum width=1cm, minimum height=2cm] (b)
[right=2 cm of a]{};

\draw[causalLink, bend left] (a-eff-1) to (b-pre-1);
\draw[causalLink, bend left, causal mark=*] (a-eff-2) to (b-pre-2);
\draw[causalLink, bend left, causal mark=@] (a-eff-3) to (b-pre-3);
\draw[causalLink, bend left, causal mark=^] (a-eff-4) to (b-pre-4);
\draw[causalLink, bend left, causal mark={&}] (a-eff-5) to (b-pre-5);
\draw[causalLink, bend left, causal mark=-] (a-eff-6) to (b-pre-6);
```



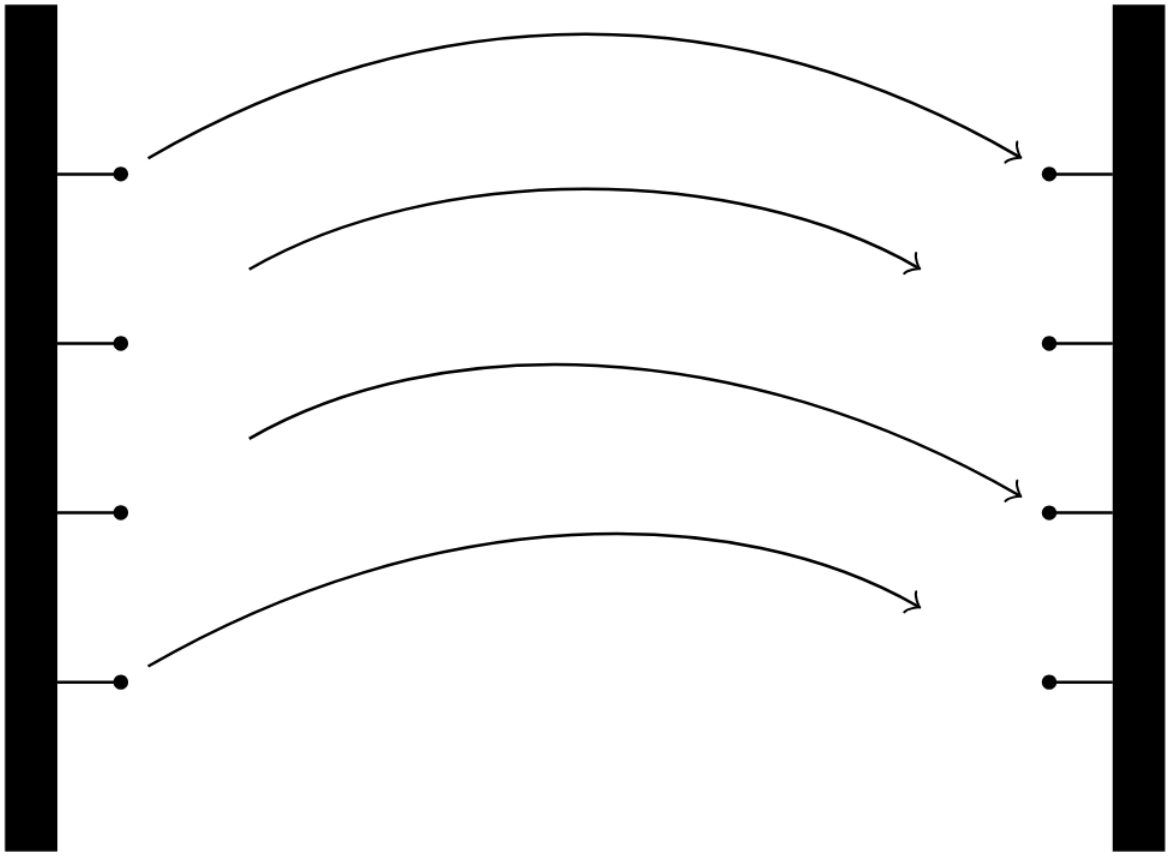
The triangle mark (`^`) automatically rotates to point along the link direction in both horizontal and vertical diagrams.



The `shorten <=<dim>, shorten >=<dim>` could used for adjust gap of causual link with start node and end node. `<=` for the start node gap, and `=>` for the end node gap:

```
\node[init={,,,}, minimum height=4cm] (init) {};
\node[goal={,,,},minimum height=4cm] (goal) [right=5cm of init] {};

\draw[causalLink, bend left] (init-eff-1)  to (goal-pre-1);
\draw[causalLink, bend left, shorten <=20pt, shorten >=20pt] (init-eff-2)  to
(goal-pre-2);
\draw[causalLink, bend left, shorten <=20pt] (init-eff-3)  to (goal-pre-3);
\draw[causalLink, bend left, shorten >=20pt] (init-eff-4)  to (goal-pre-4);
```



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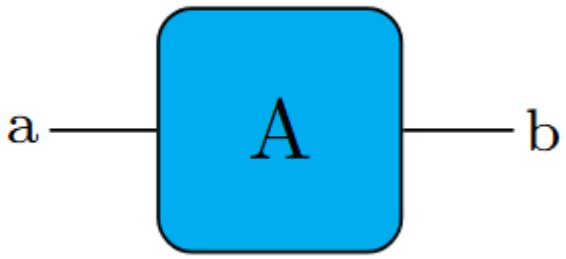
6. Label position — `label side`

By default labels sit at the **midpoint** of their line.

Add `label side` to move labels to the **tip** of the line (the far end from the node):

- Horizontal prec labels → placed at the left tip, right-aligned
- Horizontal eff labels → placed at the right tip, left-aligned
- Vertical labels → placed at the top/bottom tip with direction-aware anchoring

```
\node[action={a}{b}{A}, minimum width=1cm, minimum height=1cm,label side] (a) {};
```



`label side` could set inside a shared style so it applies to every node at once:

```
\tikzset{
    STYLE/.style n args={3}{
        action={#1}{#2}{#3},
        minimum width=2cm,
        label side
    }
}
```

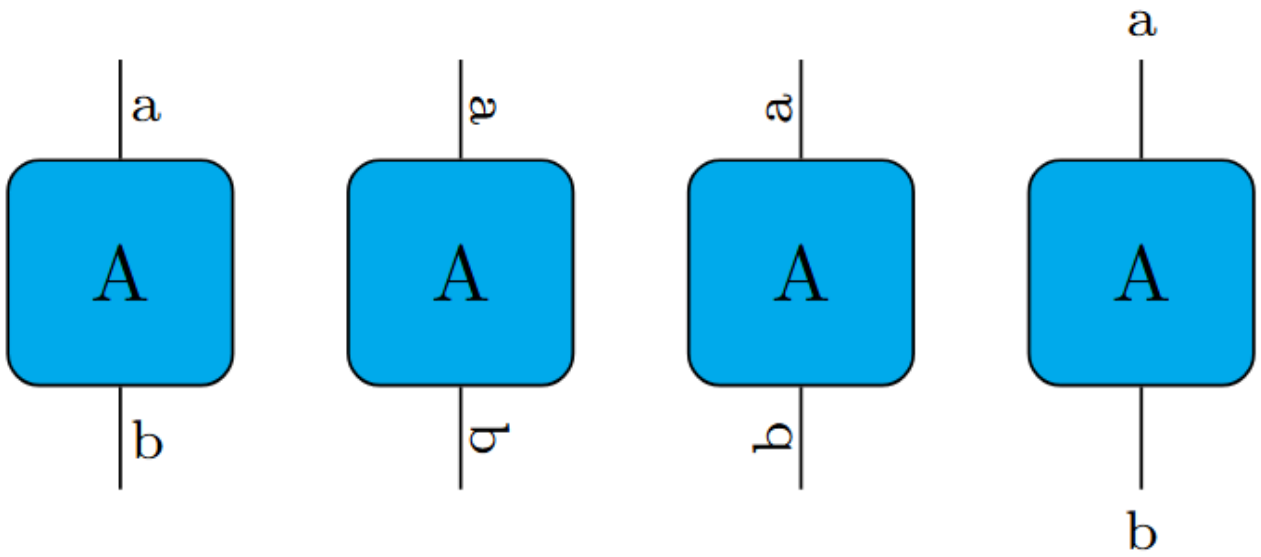
---

7. Vertical label orientation

These keys only affect **vertical-mode** label nodes. They control how label text is rotated/positioned relative to the line.

Key	Orientation
(default)	Horizontal text, to the right of the line midpoint
`vertical text`	Rotated −90°, text reads top-to-bottom along the right of the line
`vertical text right`	Same as `vertical text`
`vertical text left`	Rotated +90°, text reads bottom-to-top along the left of the line
`vertical text top`	Horizontal text, placed at the line end (above prec tips / below eff tips)

```
\node[action={a}{b}{A}, minimum width=1cm, minimum height=1cm, vertical] (a) {};
\node[action={a}{b}{A}, minimum width=1cm, minimum height=1cm, vertical,vertical
text] (b) [right=0.5cm of a] {};
\node[action={a}{b}{A}, minimum width=1cm, minimum height=1cm, vertical,vertical
text left] (c) [right=0.5cm of b] {};
\node[action={a}{b}{A}, minimum width=1cm, minimum height=1cm, vertical,vertical
text top, vertical text top shift=8pt] (d) [right=0.5cm of c] {};
```



Combine with shared style to place the text at any of these orientations:

```
\tikzset{
    VSTYLE/.style n args={3}{
        action={#1}{#2}{#3},
        vertical, minimum width=2cm,
        vertical text left
    }
}
```

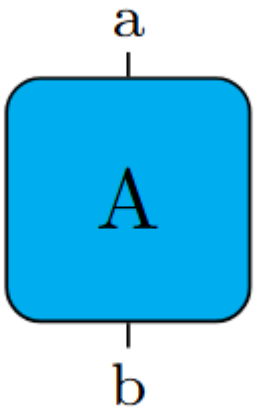
Adjusting the `vertical text top` gap

When using `vertical text top`, the label is placed a small distance from the line end. The default shift is `3pt`. Control that gap with:

```
vertical text top shift=<dim>
```

Example — decrease the gap to 2pt combine with 0.1cm label length on pre and eff:

```
\node[action={a}{b}{A}, minimum width=1cm, minimum height=1cm, vertical,vertical
text top, vertical text top shift=2pt, prec label length=0.1cm, eff label
length=0.1cm] (a) {};
```



Or set it globally for all nodes in the picture:

```
\begin{tikzpicture}[vertical text top, vertical text top shift=6pt]
    ...
\end{tikzpicture}
```

---

8. Label line length

Automatic sizing (default)

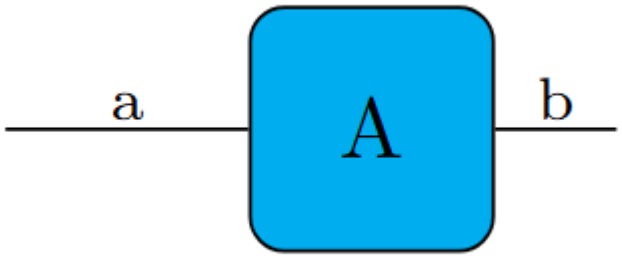
The library measures the widest label text among all prec (or eff) labels for a node and sets all lines of that group to the same length (widest text width + 0.3 cm).

Override all prec lines to a fixed length

```
prec label length=<dim>     % applies to action precs AND goal precs
eff label length=<dim>      % applies to action effs  AND init effs
```

For example:

```
\node[action={a}{b}{A}, minimum width=1cm, minimum height=1cm, prec label
length=1cm, eff label length=0.5cm] (a) {};
```



For `init` / `goal` nodes the unified keys apply too:

- `init` and `init vertical` → use `eff label length`
- `goal` and `goal vertical` → use `prec label length`

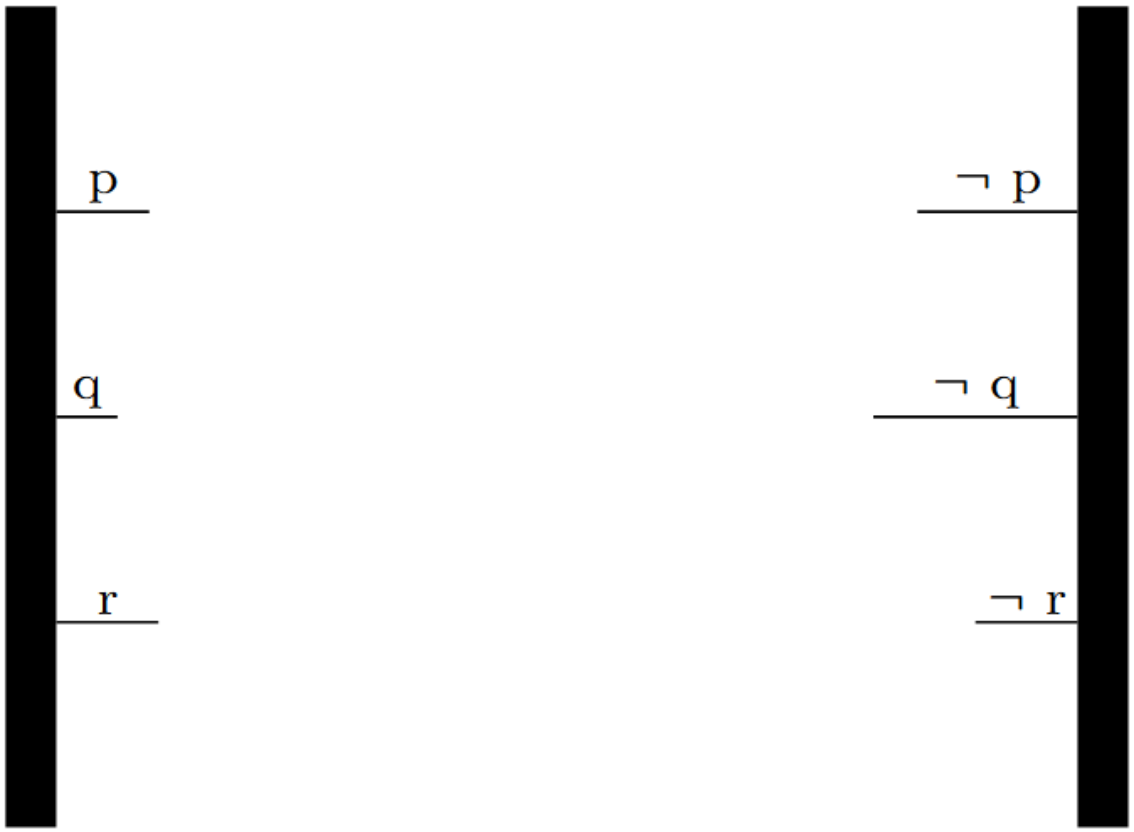
Override individual lines (per-line lengths)

Supply a comma-separated list; leave entries empty to keep auto-sizing for those positions:

```
prec label lengths={<dim1>,<dim2>,...}
eff label lengths={<dim1>,<dim2>,...}
```

Examples:

```
\node[init={p,q,r}, minimum height=4cm, eff label lengths ={,0.3cm,0.5cm}] (init)
{};
\node[goal={$\neg$ p,$\neg$ q,$\neg$ r},minimum height=4cm, prec label lengths =
{,1cm,0.5cm}] (goal) [right=5cm of init] {};
```



9. Complete style example — horizontal diagram

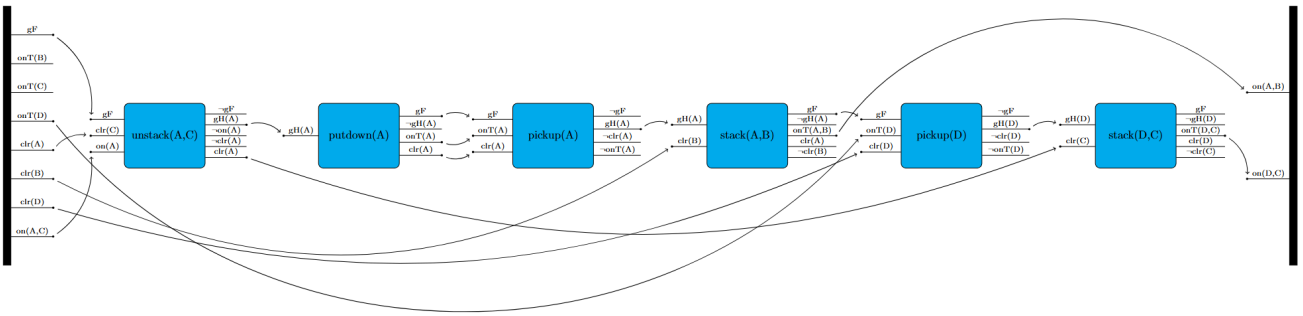
```
% Define commonly used style for action
\tikzset{
    STYLE/.style n args = {3}{
        action={#1}{#2}{#3},
        minimum height=2cm,
        minimum width=2.5cm
    }
}

% Define actions
\tikzset{
    % init and goal:
    INIT/.style = {
        init={gF,onT(B),onT(C),onT(D),clr(A),clr(B),clr(D),{on(A,C)}},
        minimum height=8cm
        },
    GOAL/.style = {
        goal={{on(A,B)},{on(D,C)}},
        minimum height=8cm
        },
    % normal actions:
    PICKUP/.style n args  ={1}{
        STYLE={gF,onT(#1),clr(#1)}{$\neg$gF,gH(#1),$\neg$clr(#1),$\neg$onT(#1)}
```

```
{pickup(#1)}
    },
    PUTDOWN/.style n args ={1}{
        STYLE={gH(#1)}{gF,$\neg$gH(#1),onT(#1),clr(#1)}{putdown(#1)}
        },
    STACK/.style n args   ={2}{
        STYLE={gH(#1),clr(#2)}{gF,$\neg$gH(#1),onT{(#1,#2)},clr(#1),$\neg$clr(#2)}
{stack{(#1,#2)}}
        },
    UNSTACK/.style n args ={2}{
        STYLE={gF,clr(#2),on(#1)}
{$\neg$gF,gH{(#1)},$\neg$on(#1),$\neg$clr(#1),clr(#1)}{unstack{(#1,#2)}}
        }
}

\node[INIT]           (init) {};
\node[UNSTACK={A}{C}] (unstackAC) [right=3.5cm of init]      {};
\node[PUTDOWN={A}]    (putdownA)  [right=3.5cm of unstackAC] {};
\node[PICKUP={A}]     (pickupA)   [right=3.5cm of putdownA]  {};
\node[STACK={A}{B}]   (stackAB)   [right=3.5cm of pickupA]   {};
\node[PICKUP={D}]     (pickupD)   [right=3.5cm of stackAB]   {};
\node[STACK={D}{C}]   (stackDC)   [right=3.5cm of pickupD]   {};
\node[GOAL]           (goal)      [right=3.5cm of stackDC]   {};

\draw [causalLink, bend left]     (init-eff-1)      to (unstackAC-pre-1);
\draw [causalLink, bend left]     (init-eff-5)      to (unstackAC-pre-2);
\draw [causalLink, bend right]    (init-eff-8)      to (unstackAC-pre-3);
\draw [causalLink, bend left]     (unstackAC-eff-2) to (putdownA-pre-1);
\draw [causalLink, bend left]     (putdownA-eff-1)  to (pickupA-pre-1);
\draw [causalLink, bend right]    (putdownA-eff-3)  to (pickupA-pre-2);
\draw [causalLink, bend right]    (putdownA-eff-4)  to (pickupA-pre-3);
\draw [causalLink, bend left]     (pickupA-eff-2)   to (stackAB-pre-1);
\draw [causalLink, bend right]    (init-eff-6)      to (stackAB-pre-2);
\draw [causalLink, bend left]     (stackAB-eff-1)   to (pickupD-pre-1);
\draw [causalLink, bend right=50]     (init-eff-4)      to (pickupD-pre-2);
\draw [causalLink, bend right=20] (init-eff-7)      to (pickupD-pre-3);
\draw [causalLink, bend left]     (pickupD-eff-2)   to (stackDC-pre-1);
\draw [causalLink, bend right=20] (unstackAC-eff-5) to (stackDC-pre-2);
\draw [causalLink, bend left=50]     (stackAB-eff-3)   to (goal-pre-1);
\draw [causalLink, bend left]     (stackDC-eff-3)   to (goal-pre-2);
```



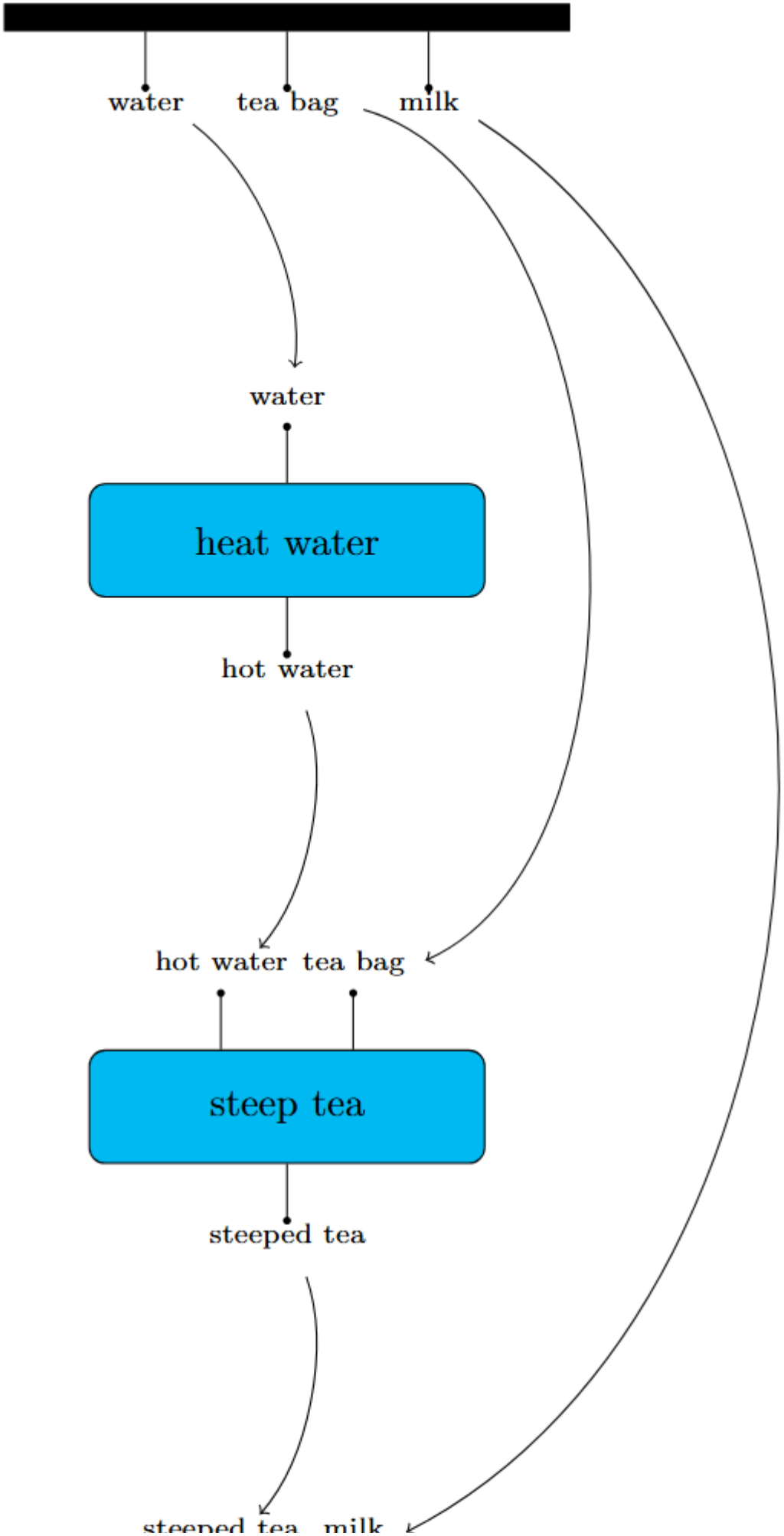
10. Complete style example — vertical diagram

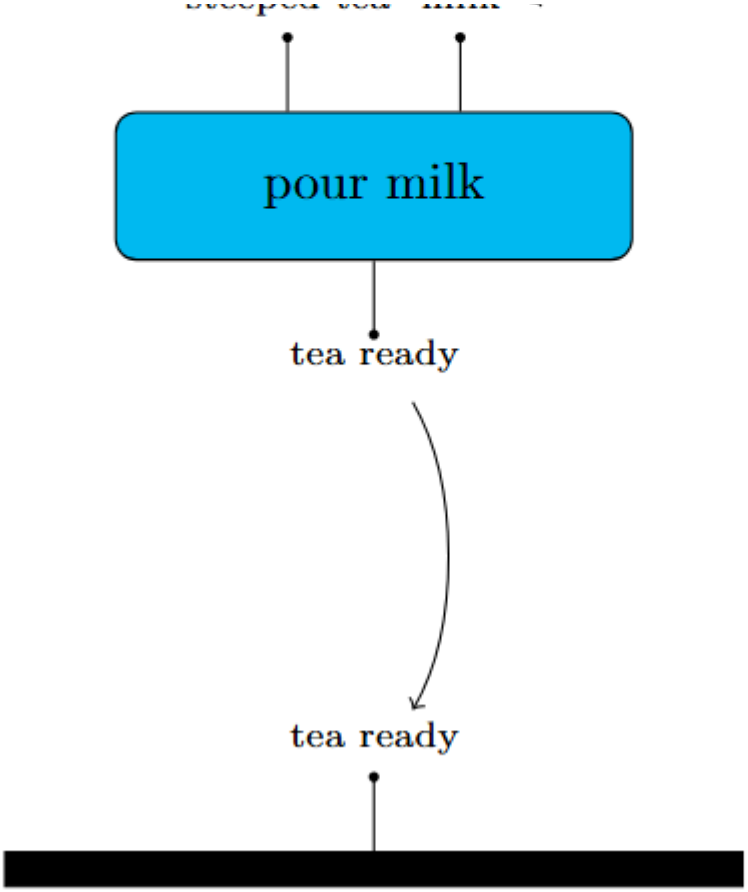
```
\tikzset{
    VSTYLE/.style n args={3}{
        action={#1}{#2}{#3},
        vertical, minimum width=3.5cm, minimum height=1cm,
        vertical text top,
        vertical text top shift=0.25cm,
        prec label length={0.5cm},
        eff label length={0.5cm},
    },
    VINIT/.style={
        init vertical={water,tea bag,milk},
        minimum width=5cm, minimum height=0.1cm,
        vertical text top,
        vertical text top shift=0.25cm,
        eff label length={0.5cm}
    },
    VGOAL/.style={
        goal vertical={tea ready},
        minimum width=5cm, minimum height=0.03cm,
        vertical text top,
        vertical text top shift=0.2cm,
        prec label length={0.5cm}
    },
    HEAT/.style={
        VSTYLE={water}{hot water}{heat water}
        },
    STEEP/.style={
        VSTYLE={{hot water},{tea bag}}{steeped tea}{steep tea}
        },
    POUR/.style={
        VSTYLE={{steeped tea},milk}{tea ready}{pour milk}
        }
}

\node[VINIT]  (init)  {};
\node[HEAT]   (heat)  [below=4cm of init]  {};
\node[STEEP]  (steep) [below=4cm of heat]  {};
\node[POUR]   (pour)  [below=4cm of steep] {};
\node[VGOAL]  (goal)  [below=4cm of pour]  {};

\draw[causalLink, bend left, shorten <=10pt, shorten >=10pt] (init-eff-1)  to
(heat-pre-1);
\draw[causalLink,bend left=70, shorten <=20pt, shorten >=20pt] (init-eff-2)  to
(steep-pre-2);
\draw[causalLink,bend left=60, shorten <=10pt, shorten >=10pt] (init-eff-3)  to
(pour-pre-2);
\draw[causalLink,bend left, shorten <=10pt, shorten >=10pt] (heat-eff-1)  to
(steep-pre-1);
\draw[causalLink,bend left, shorten <=10pt, shorten >=10pt] (steep-eff-1) to
(pour-pre-1);
```

```
  \draw[causalLink,bend left, shorten <=10pt, shorten >=10pt] (pour-eff-1)  to
  (goal-pre-1);
```


water
tea bag
milk
water
heat water
hot water
hot water
tea bag
steep tea
steeped tea
steeped tea
milk



---

11. Key reference summary

Node styles

Style	Arguments	Description
`action`	`{precs}{effs}{name}`	Horizontal or vertical action node
`init`	`{labels}`	Vertical bar, effects go right
`goal`	`{labels}`	Vertical bar, precs go left
`init vertical`	`{labels}`	Horizontal bar, effects go down
`goal vertical`	`{labels}`	Horizontal bar, precs go up

Layout and label keys

Key	Effect
`horizontal`	Precs left, effs right *(default)*
`vertical`	Precs up, effs down
`label side`	Labels at line tips instead of midpoints
`vertical text` / `vertical text right`	Labels rotated −90° (top-to-bottom)

Key	Effect
`vertical text left`	Labels rotated +90° (bottom-to-top)
`vertical text top`	Labels horizontal at line ends
`vertical text top shift=<dim>`	Gap between label and line end (default `3pt`)

Line length keys

Key	Effect
`prec label length=<dim>`	Fixed length for all prec lines (action + goal)
`eff label length=<dim>`	Fixed length for all eff lines (action + init)
`prec label lengths={d1,d2,...}`	Per-line lengths for prec lines; empty entry = auto
`eff label lengths={d1,d2,...}`	Per-line lengths for eff lines; empty entry = auto

Causal link keys

Key	Effect
`causalLink`	Arrow with endpoint marks
`causalLink with label={text}`	Arrow with mid-path label
`causal mark=-`	No endpoint mark
`causal mark=@`	Hollow circle mark
`causal mark=*`	Star mark
`causal mark=^`	Triangle mark
`causal mark={&}`	Square mark

Named anchor coordinates

Coordinate	Where
`(nodename-pre-N)`	Tip of the N-th precondition line
`(nodename-eff-N)`	Tip of the N-th effect line